

Rebuttal

----- REVIEW 1 -----

C1.1: The paper suffers from concerns regarding dataset construction and potential selection bias, as a significant portion of migration records (101 out of 301) are excluded without sufficient analysis of their differences from retained cases.

R1.1: Thank you for the comment. We have analyzed the 101 excluded records and added the following paragraph to the revised manuscript.

"For the 101 excluded records, we further investigated their failure reasons. Among them, 51 failures were caused by dependency changes, including failures to download dependencies and test failures triggered by changes in dependent packages. In addition, 35 failures were due to missing secrets unavailable in our environment, 8 were caused by outdated externally linked resources, and 7 resulted from unsupported runtime environments in GitHub Actions."

We also compared the configuration size and the number of migration commits between the excluded and retained records. For the 101 excluded records, the average number of lines in the original Travis CI configurations was 21, while the average number of lines in the target GitHub Actions configurations was 39. Among these records, 40 required multiple commits to complete the migration. For the retained 200 records, the corresponding values were 20, 39, and 106, respectively. These results suggest that there is no substantial difference between the excluded and retained records

C1.2: The notion of configuration equivalence and taxonomy construction relies heavily on manual judgment, yet lacks sufficient details such as criteria, inter-rater agreement, and reproducibility evidence.

R1.2: Thank you. We agree that both tasks rely heavily on manual judgement. To ensure reliability, we followed an open-coding procedure~\cite{Seaman1999QME}. The annotation was independently performed by two authors, one with seven years of CI experience and the other with three years. In cases of disagreement, a third author with five years of CI experience was consulted to resolve conflicts and reach a consensus.

For configuration equivalence, we followed explicit rules to process the translation records. We kept the source Travis CI configurations unchanged to preserve the

original migration context. If a migrated GitHub Actions configuration introduced functionality not specified in the original Travis CI configuration or not supported by Travis CI by default, we removed that functionality. If it omitted functionality specified in Travis CI, we removed the corresponding record, since manually adding the missing functionality could produce configurations that differ from what developers would actually write. The two authors achieved substantial agreement in this process, with a Cohen's Kappa of 0.78.

For taxonomy construction, the two annotators independently identified translation issues from the generated configurations, execution logs, and related documentation. They then grouped similar issues into initial labels, refined those labels through discussion, and organized them into the high-level categories reported in the paper. To assess the reliability of the coding process, we measured inter-rater agreement and obtained a Cohen's Kappa of 0.77, indicating substantial agreement.

C1.3: Furthermore, the effort analysis uses only indirect proxies, and the validity of using the same dataset for both guideline derivation and evaluation is unclear, raising concerns about potential evaluation bias and generalizability.

R1.3: Thank you. We agree that the most direct way to measure migration effort would be to investigate developers' actual time spent or cognitive workload. However, such information is not available from repository histories and cannot be reliably reproduced. Therefore, we rely on indirect metrics, including configuration size and migration attempts, to characterize observable migration effort. These metrics are inspired by the intuition that larger configuration changes and more migration commits usually indicate greater migration effort.

We agree that using the same set of configurations to derive the issue taxonomy and evaluate the guideline-based prompting strategy may introduce a circular dependency. To address this concern, we have added several sentences to the threats-to-validity section as follows.

"Finally, another threat concerns our guideline-based prompting strategy. The guidelines were derived from 200 translation records and evaluated on the same dataset, which may introduce a circular dependency and potentially overestimate their effectiveness. To mitigate this threat, we designed the guidelines to be general rather than configuration-specific, such as ensuring valid YAML syntax and representing version numbers as strings. We believe that CI translation rules are relatively limited in scope, and these 200 translation records provide a reasonable basis for covering

common translation issues. In addition, the guideline-based strategy is extensible, as new rules can be added when new issue types are observed."

C1.4: 1. The notion of configuration equivalence appears to rely on substantial manual judgment. Since a large proportion of candidate migration records were discarded during this process, it would be helpful to provide more detailed criteria, examples, and inter-rater statistics specifically for the equivalence assessment to improve reproducibility and reduce concerns about selection bias.

R1.4: Thank you for the comment. We have addressed this concern in R1.2.

C1.5: 2. The final evaluation is conducted on only 200 executable migration records, while 101 records are excluded. (in RQ2 and RQ3) It remains unclear whether these excluded cases systematically differ from the retained ones (e.g., being more complex or harder to migrate). Such filtering may introduce selection bias and potentially overestimate the effectiveness of the studied approaches. Additional analysis comparing retained and excluded cases would strengthen the validity of the conclusions.

R1.5: Thank you for the comment. We have addressed this concern in R1.1.

C1.6: 3. The guideline-based prompting strategy is derived from the issue taxonomy identified in RQ2. It is unclear whether the same evaluation dataset was used both to construct the guidelines and to evaluate their effectiveness. If so, there may be a risk of evaluation bias. Clarification regarding dataset separation would be valuable.

R1.6: Thank you for the comment. We have addressed this concern in R1.3.

C1.7: 4. The effort analysis mainly relies on configuration size and commit statistics, which may not fully reflect the actual migration effort. The authors should better justify these metrics as proxies for effort and discuss potential limitations.

R1.7: Thank you for the comment. We have addressed this concern in R1.3.

C1.8: 5. The proposed issue taxonomy is valuable, but the paper does not clearly

describe how it was constructed and validated. More details on the labeling process and annotation agreement would strengthen the credibility of the taxonomy.

R1.8: Thank you for the comment. We have addressed this concern in R1.2.

C1.9: 6. The study focuses exclusively on Travis CI to GitHub Actions migration. While this migration path is practically important, it remains unclear whether the reported findings generalize to other CI/CD platforms with different configuration paradigms and migration characteristics. A discussion of external validity or additional evaluation on other migration scenarios would strengthen the paper.

R1.9: Thank you for the comment. We acknowledge that the original threats-to-validity section only briefly discussed the limitations of our migration scenario and did not provide sufficient detail. To address this issue, we have revised this part and added a more comprehensive discussion, as follows:

" First, our study focuses exclusively on migrations from Travis CI to GitHub Actions, motivated by the practical prevalence of this migration pattern. We cannot guarantee that our results generalize to other CI platforms or migration scenarios. However, we believe that some of the findings are broadly applicable. CI migration often requires substantial effort beyond the specific case of Travis CI to GitHub Actions, because CI tools typically provide extensive documentation, adopt platform-specific syntax, and exhibit semantic differences in build environments, triggers, dependency management, caching, and deployment mechanisms. In addition, some issue types in our taxonomy may also occur in other CI migration scenarios. For example, since many CI platforms use YAML-based configurations, syntax errors and incorrect value formats may also appear in migrations between other CI platforms. Moreover, our enhancement strategies are largely platform-agnostic. Iterative refinement can be applied whenever execution feedback is available, and guideline-based prompting can be adapted to other platforms by replacing or extending the platform-specific rules."

----- REVIEW 2 -----

C2.1: -The study is limited to migration from Travis CI to GitHub Actions and only Python/JavaScript projects. Also, it seems that the use of Travis CI represents a small portion of the CI tools used by the organizations (<https://blog.jetbrains.com/teamcity/2025/10/the-state-of-cicd/>).

R2.1: Thank you for raising this concern. We agree that our study is limited to Travis CI-to-GitHub Actions migration and Python/JavaScript projects, which may affect generalizability. We selected this setting because Travis CI was once widely used in open-source projects and has increasingly been replaced by GitHub Actions, providing a practical migration scenario for studying CI configuration translation. We have clarified this limitation in the revised manuscript and added future work to extend the study to more programming languages and heterogeneous CI/CD systems.

C2.2: -The paper does not use the template for the conference. It uses an ACM template instead.

R2.2: Thank you for the comment. We will revise the manuscript using the correct conference template.

C2.3: -The dataset is reduced from 301 migration records to 200 executable records for RQ2/RQ3, and the effect of excluding the other records should be discussed more clearly.

R2.3: Thank you for the comment. Reviewer 1 raised a similar concern, and we repeat our response here. We have analyzed the 101 excluded records and added the following paragraph to the revised manuscript.

"For the 101 excluded records, we further investigated their failure reasons. Among them, 51 failures were caused by dependency changes, including failures to download dependencies and test failures triggered by changes in dependent packages. In addition, 35 failures were due to missing secrets unavailable in our environment, 8 were caused by outdated externally linked resources, and 7 resulted from unsupported runtime environments in GitHub Actions."

We also compared the configuration size and the number of migration commits between the excluded and retained records. For the 101 excluded records, the average number of lines in the original Travis CI configurations was 21, while the average number of lines in the target GitHub Actions configurations was 39. Among these records, 40 required multiple commits to complete the migration. For the retained 200 records, the corresponding values were 20, 39, and 106, respectively. These results suggest that there is no substantial difference between the excluded and retained records

C2.4:-Why are 101 of the 301 migration records excluded from RQ2/RQ3? Could excluding non-executable records bias the issue distribution and enhancement results?

R2.4: Thank you for the comment. We have addressed this concern in R2.3.

C2.5:-Can you provide more detail on the log matching process and its failure modes? Logs can differ even when workflows are semantically equivalent.

R2.5: We rely on ActionRemaker to match execution logs. It ignore unstable information, such as timestamps, and then extracts key attributes, such as build outcomes, test execution status, and test counts. The normalized attributes are then compared, and two logs are considered matched only when all key attributes are equivalent.

C2.6:-For Guideline+IR, how many iterations are needed on average, and what is the cost in tokens and time?

R2.6: Thank you for the comment. On average, Guideline+IR requires 1.59 iterations per translation, with each request consuming approximately 1,437 tokens and taking 3.9 seconds. In addition, we report the detailed costs for different LLMs and prompting strategies in Tables 2.1, 2.2, and 2.3.

Table 2.1 Average Cost of Basic Prompt Across Models

Models	Gemini3	GPT-4o	GPT-4o-mini	Llama	Qwen3	DeepSeek
Average tokens	374	367	403	347	2437	538
Average time	3.52	3.43	5.07	2.63	44.38	4.50

Table 2.2 Average Cost of Guideline-based Prompt Across Models

Models	Gemini3	GPT-4o	GPT-4o-mini	Llama	Qwen3	DeepSeek
Average tokens	856	825	803	786	2835	1051
Average time	3.10	3.51	4.87	2.74	45.93	3.45

Table 2.3 Average Cost of Gemini 3 Flash Across Different Prompting Strategies

Strategies	Oneshot	IR	IR+Guideline
Average tokens	707	1021	1437
Average time	4.90	3.73	3.90

C2.7: -How many LLMs are actually evaluated? The text mentions six representative models but lists more names, including GPT-5.1 code. However, table 1 does not mention this model.

-The model list and result tables should be made consistent.

R2.7: We evaluated six LLMs in total. “GPT-5.1 code” was mistakenly mentioned. We have removed it.

C2.8: -It would help to include more examples of difficult migrations, especially those involving secrets, matrix builds, and deployment steps.

R2.8: Thank you for the suggestion. Due to space limitations, we cannot include many such examples in the paper, but we have added representative cases to our companion website.

C2.9: -Figure 1 is hard to read. Can you improve the visuals of it?

R2.9: Thank you for the comment. We have redrawn Figure 1 to improve its readability. In the revised figure, we adjusted the layout and added more detailed comments to clarify the mapping between Travis CI and GitHub Actions configurations. We would be happy to further revise the figure if there are any additional suggestions.

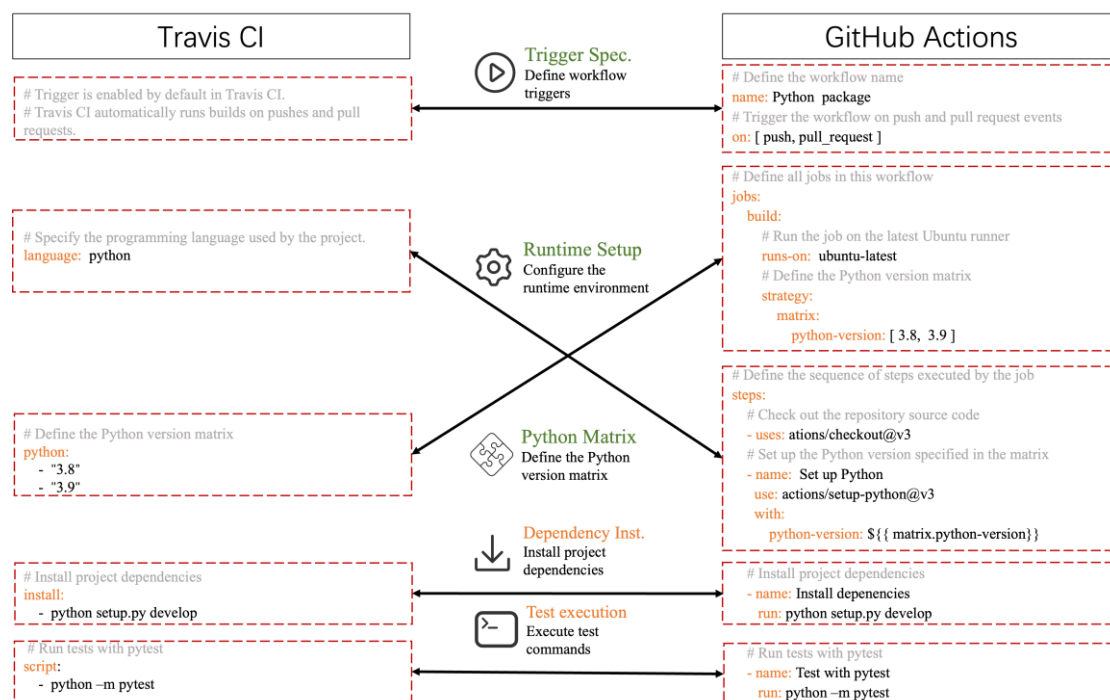


Figure 2.1 Example of Equivalent Travis CI and GitHub Actions Configurations

C2.10: -A figure to explain the methodology would help understanding more the study.

R2.10:Thank you for the suggestion. We have added a new methodology overview figure to the revised manuscript. The figure illustrates the overall workflow of our study, including data preparation, migration effort analysis, issue taxonomy construction, and enhancement strategy evaluation, thereby improving the readability and understandability of the methodology.

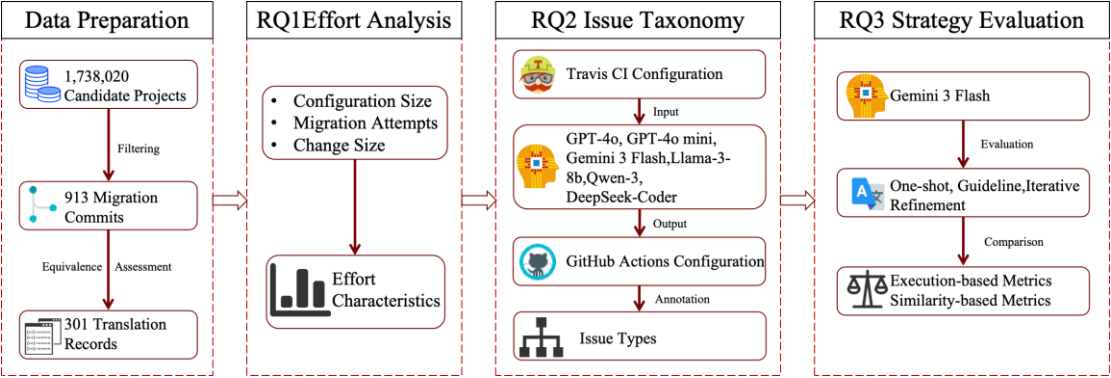


Figure 2.2 Overview of Our Methodology

C2.11:-The charts could have been added as PDF to improve the legibility.

R2.11: Thank you for the suggestion. We have regenerated the charts in PDF format.

C2.12: -Future work could have been provided.

R2.12: Thank you for the suggestion. We have added a discussion of future work to the conclusion section. Specifically, we now discuss future directions such as exploring more advanced LLM-based techniques for CI configuration translation and extending the current Travis CI-to-GitHub Actions migration scenario to a more general setting involving multiple heterogeneous CI/CD systems.

“In future work, we plan to explore more advanced LLM-based techniques for CI configuration translation. We also plan to extend the studied scenario from Travis CI-to-GitHub Actions migration to a more general setting involving multiple heterogeneous CI/CD systems.”

C2.13: TYPOS:

- "becauseas" in the threats-to-validity section.

- 75 casses -> 75 cases (section 4.2.2)

R2.13: We have corrected the typos.

----- REVIEW 3 -----

C3.1: - Key numerical claims in the Introduction and Abstract directly contradict Table 3 and Figure 4b.

- Section 3.3 and Table 1 contain inconsistent and unverifiable model information.
- RQ3 evaluates all enhancement strategies on one model but presents model-agnostic conclusions.
- The guideline strategy was derived from and evaluated on the same configurations, creating a circular dependency.
- Section 4.2 misidentifies Llama-3-8B as Qwen-3 in the model ranking sentence.
- Section 4.3 reports an incorrect BSR for one-shot prompting that contradicts Table 3.
- Section 3.3 describes Qwen-3 inference with two contradictory settings.
- The 101 configurations excluded for non-executability have no documented failure breakdown.
- ROUGE-L is reported as the maximum of precision, recall, and F1, which is non-standard.
- No statistical significance tests are reported for any performance comparison.
- The replication package README contains multiple reproducibility gaps.

R3.1: Thank you for summarizing these major concerns. Since the following comments provide detailed explanations of these issues, we respond to each point individually in the corresponding responses below.

C3.2: 1. The Introduction and Abstract contain numerical results that are inconsistent with the actual experimental data. The Introduction states "approximately 30.9% of translations require more than one commit," but Section 4.1 and Figure 4b show that 146 out of 301 records required multiple commits, which is 48.5%. More critically, the

RQ3 summary in the Introduction claims "a BSR of 75.5% (173 successful translations), representing nearly a threefold improvement over the basic LLM baseline of 25.8%." Table 3 shows the combined method achieves 88.5% (177) with a basic BSR of 63.0% (126). The authors should align all introductory and abstract claims with the values reported in Table 3 and Figure 4b.

R3.2: Thank you for identifying these inconsistencies. During the experiments and manuscript preparation, we revised the results several times, and some outdated values were not fully updated in the earlier version of the manuscript. We have carefully rechecked the results and ensured that the numerical values are consistent across the Abstract, Introduction, Section 4.1, Figure 4b, and Table 3. Specifically, we have revised the migration-attempt result to state that 48.5% of translations require more than one commit. We have also corrected the RQ3 summary as follows: the combined strategy of guideline-based prompting and iterative refinement achieves the best performance, with a BSR of 88.5% (177 successful translations), representing a 25.5 percentage-point improvement over the basic LLM baseline of 63.0%.

C3.3: 2. The model description in Section 3.3 and Table 1 contains several issues. First, Section 3.3 names seven models while stating six models were studied; "GPT-5.1 code" carries reference [4] (OpenAI) but does not appear in Table 1 or any subsequent analysis. Second, Table 1 lists Gemini 3 Flash with a release date of 2025-12-17, which is after the August 2025 retrieval timestamps cited in the references and therefore inconsistent with the stated experimental timeline; the authors should confirm the exact Gemini model version and API snapshot that was used. The authors should remove "GPT-5.1 code" from Section 3.3, verify all release dates, and ensure the model list in the text matches Table 1 exactly.

R3.3: Thank you for pointing out these inconsistencies in the model description. We have removed "GPT-5.1 code" from Section 3.3 and verified the model information reported in Table 1.

C3.4 3. RQ3 evaluates all three enhancement strategies on Gemini 3 Flash only. The paper justifies this by noting Gemini 3 Flash was the best model in RQ2, but that does not establish that the relative benefit of each strategy holds across models of different capabilities. For the iterative refinement strategy, running all six models would require a large number of GitHub Actions workflow executions across multiple iterations, which represents a genuine cost and time constraint. For the guideline strategy, no such

constraint applies: guideline prompting adds only a text block to the prompt and incurs no additional execution cost. The relative improvement from guideline prompting may vary substantially across models. Notably, Llama-3-8b committed 145 Unspecified Default errors in RQ2, and the corresponding guideline rule may improve its BSR by a much larger margin than for Gemini 3 Flash, which committed only 14 such errors. The authors should run the guideline strategy on all six models and report the results; for the iterative refinement strategy, scoping the evaluation to Gemini 3 Flash is acceptable provided all model-agnostic claims about IR are removed and conclusions are explicitly limited to Gemini 3 Flash.

R3.4:

We thank the reviewer for this suggestion. Following the reviewer's recommendation, we applied the guideline strategy to all six evaluated models and have included the results in the revised manuscript. We observe performance improvements for five of the six models, indicating that the effectiveness of guideline prompting is generally consistent across different models, although the magnitude of improvement varies.

DeepSeek is the only exception. Although DeepSeek is specifically fine-tuned for coding tasks and our Travis CI-to-GitHub Actions migration task is code-oriented, the guideline strategy does not improve its performance. One possible explanation is that the additional guideline prompts interfere with task-specific generation patterns learned during code-focused training.

Table 3.1 Performance of LLMs under Guideline-based Prompting

Model	BSR	LMR
Gemini3	63.0 (126)	96.0 (121)
Gemini3-Guideline	71.5 (143)	95.8 (137)
GPT-4o	34.5 (69)	92.8(64)
GPT-4o-Guideline	54.5 (109)	88.1 (96)
GPT-4o-mini	34.0 (68)	92.6 (63)
GPT-4o-mini-Guideline	47.5 (95)	89.5 (85)
Llama	19.5 (39)	92.3 (36)
Llama-Guideline	36.0 (72)	90.3 (65)
Qwen3	26.0 (52)	88.5 (46)
Qwen3-Guideline	39.0 (78)	88.5 (69)
DeepSeek	18.5 (37)	78.4 (29)
DeepSeek-Guideline	18.0 (36)	72.2 (26)

C3.5 4. The guideline-based prompting strategy was designed by converting the RQ2 issue taxonomy into natural language rules, and the same 200 configurations were used both to derive the taxonomy and to evaluate whether the guidelines improve translation quality. This creates a circular dependency in which the guidelines are implicitly fitted to the test set: the annotators identified which specific errors appeared in these 200 configurations, and the guidelines were written to address exactly those errors. As a result, the reported 71.5% BSR for the guideline strategy may overestimate its ability to generalise to unseen configurations. The authors should add a paragraph to the threats to validity section acknowledging this circular dependency, qualify the 71.5% BSR result as a potential overestimate rather than a generalisation measure, and clarify whether each guideline rule was written to address an error type in general or a configuration-specific pattern observed in these 200 cases.

R3.5: Thank you for raising this important concern. We agree that using the same set of configurations to derive the issue taxonomy and evaluate the guideline-based prompting strategy may introduce a circular dependency. To address this concern, we have added several sentences to the threats-to-validity section as follows.

"Finally, another threat concerns our guideline-based prompting strategy. The guidelines were derived from 200 translation records and evaluated on the same dataset, which may introduce a circular dependency and potentially overestimate their effectiveness. To mitigate this threat, we designed the guidelines to be general rather than configuration-specific, such as ensuring valid YAML syntax and representing version numbers as strings. We believe that CI translation rules are relatively limited in scope, and these 200 translation records provide a reasonable basis for covering common translation issues. In addition, the guideline-based strategy is extensible, as new rules can be added when new issue types are observed."

C3.6: 1. Section 4.2 states "Gemini 3 Flash exhibited the fewest issues (75 cases), followed by GPT-4o (136), GPT-4o mini (154), Qwen-3 (172), DeepSeek-Coder (174), and Qwen-3 (274)", where the final entry should be "Llama-3-8B (274)."

R3.6: We have corrected the model name.

C3.7: 2. Section 4.3 states "One-shot prompting reduced performance to 63.0% (114 successful translations)." However, $114/200 = 57.0\%$, not 63.0%.

R3.7: Thank you for pointing this out. We have corrected the One-shot prompting result in Section 4.3 from "63.0% (114 successful translations)" to "57.0% (114 successful

translations)."

C3.8: 3. Section 3.3 describes the inference configuration for Qwen-3 inconsistently. One sentence states "For Llama-3-8B and Qwen-3, inference was performed using greedy decoding, with sampling disabled (do_sample = False)," while the next sentence states "In contrast, Qwen-3 was run with its default configuration, as setting the temperature to 0 leads to infinite loops." These two descriptions contradict each other.

R3.8: Thank you for pointing this out. The first sentence contained a typo. We have corrected "For Llama-3-8B and Qwen-3" to "For Llama-3-8B and DeepSeek-Coder." Qwen-3 was evaluated using its default configuration as described in the following sentence. The manuscript has been revised accordingly.

C3.9: 4. Among the 301 migration records, 101 (33.6%) could not be executed in the current environment and were excluded from RQ2 and RQ3, yet the paper provides no breakdown of why these configurations fail. If a disproportionate number of failures are caused by missing secrets or deprecated actions, then the corresponding error categories (Missing Secret, Unsupported Action) are systematically underrepresented in the taxonomy. The authors should report the distribution of failure reasons for the 101 excluded records and discuss this as a threat to internal validity.

R3.9: Thank you for the comment. We have analyzed the 101 excluded records and added the following paragraph to the revised manuscript.

"For the 101 excluded records, we further investigated their failure reasons. Among them, 51 failures were caused by dependency changes, including failures to download dependencies and test failures triggered by changes in dependent packages. In addition, 35 failures were due to missing secrets unavailable in our environment, 8 were caused by outdated externally linked resources, and 7 resulted from unsupported runtime environments in GitHub Actions."

C3.10: 5. ROUGE-L is reported as "the maximum of precision, recall, and F1." This is non-standard: selecting the highest of three values cherry-picks the most favourable metric for each configuration. The authors should report the F1 score consistently, as done in most prior work on code translation evaluation.

R3.10: We thank the reviewer for identifying this issue. Our implementation currently reports the maximum among ROUGE-L precision, recall, and F1, following the evaluation script adopted in our codebase. We agree that reporting ROUGE-L F1 is the standard practice in code translation literature and provides a fairer basis for comparison.

We have recomputed the results using ROUGE-L F1. The overall ranking of methods and the main conclusions remain unchanged. We will update the manuscript to report ROUGE-L F1 consistently in the revised version.

C3.11:6. No statistical significance tests are reported for any performance comparison in Tables 2 or 3. The gain from IR (85.5%) to Guideline+IR (88.5%) represents only six additional successful configurations out of 200. The authors should report significance tests (e.g., McNemar's test for paired binary outcomes such as BSR) for the main pairwise comparisons, particularly IR vs. Guideline+IR and Basic vs. each enhancement strategy.

R3.11:We thank the reviewer for this suggestion. Following the reviewer's recommendation, we conducted McNemar's exact test on the paired success/failure outcomes for the major pairwise BSR comparisons.

For the comparison between IR and Guideline+IR, Guideline+IR achieved 8 unique successes on configurations where IR failed, while IR achieved 2 unique successes on configurations where Guideline+IR failed. McNemar's exact test yielded $p = 0.109$, indicating that the observed improvement does not reach the conventional 0.05 significance threshold. Basic vs oneshot: 0.126. Basic vs IR: 5.68×10^{-14} . Basic vs Guideline+IR: 8.88×10^{-16} .

We have reported the statistical significance results for all major pairwise comparisons in the revised manuscript and revised the wording of our conclusions accordingly. While the IR-to-Guideline+IR improvement should be interpreted as a positive trend rather than a statistically significant gain, Guideline+IR still achieves the highest overall BSR (88.5%) among all evaluated strategies.

C3.12:7. In the replication package README, the run command shown under "Check Build Result" is ``python -m scripts.run_iterative``, which does not match the existing ``run_CheckBuildResult.py`` script in the repository, and the step description is identical to the Iterative section, indicating a copy-paste error.

R3.12:We have corrected this error.